

标题 1: Does Green Finance Pilot Policy Foster High-Quality Green Innovation?

Evidence from China

标题 2: Green Finance and the Quality of Green Innovation: A Quasi-Natural Experiment from China's Pilot Zones

This study falls within the scope of Technology Impact and Assessment, one of the core domains of this journal. It systematically examines the impact of the green finance pilot zone policy—conceptualized as an institutional technology—on the quality of corporate green innovation, providing evaluation evidence for policymakers. Specifically, taking China's GFPZ policy as a quasi-natural experiment, we employ a multi-period difference-in-differences model to identify its causal effects on green innovation quality.

3. Research design

3.1. Construction of the core model

To empirically identify the causal effect of green finance pilot zones on corporate green innovation, we employ a multi-period difference-in-differences (DID) approach. Given that the pilot zones were established in three batches (2017, 2019, and 2022), with staggered implementation timing across regions, the multi-period DID framework is particularly suitable for causal identification. The baseline regression model is specified as follows:

$$Greenpatent_{i,t} = \alpha_0 + \alpha_1 Treat_i \times Post_{i,t} + \sum_k \beta_k Controls_{k,i,t} + \gamma_i + \mu_t + \varepsilon_{i,t} \quad (1)$$

where the subscripts i and t represent the enterprise individual and year, respectively. We mainly focus on the estimated coefficient (α_1) of the interaction term $Treat_i \times Post_{i,t}$ (policy shock interaction term, denoted as pro_{did}), which captures the net effect of the GFPZ policy on corporate green innovation. The dependent

variable $Greenpatent_{i,t}$ represents the level of green innovation for enterprise i in year. $Treat_i$ is a treatment dummy variable, defined as follows:

$$Treat_i = \begin{cases} 1, & \text{If enterprise } i \text{ is located in green finance pilot zones} \\ 0, & \text{otherwise} \end{cases} \quad (1-01)$$

Similarly, $Post_{i,t}$ is a time dummy variable, defined as:

$$Post_{i,t} = \begin{cases} 1, & \text{If year} \geq \text{establishment year of the pilot zone where enterprise } i \text{ is located} \\ 0, & \text{otherwise} \end{cases} \quad (1-02)$$

$Controls_{k,i,t}$ denotes a set of firm-level control variables. Moreover, we attempt to control for multi-dimensional fixed effects to address endogeneity concerns arising from omitted variable bias. Specifically, γ_i represents the province-level fixed effects, capturing unobserved time-invariant heterogeneity across province. μ_t denotes the year-level fixed effects, accounting for unobserved unit-invariant shocks that may affect green innovation, such as macroeconomic fluctuations and policy changes. $\varepsilon_{i,t}$ is the random error term. The coefficient of interest, (α_1) is expected to be significantly positive, indicating that the establishment of green finance pilot zones effectively promotes corporate green innovation.

3.2. Measurement of variables

Table 1 presents the definitions of the main variables involved in this study. The core variables include dependent variables, independent variables, mediating variables, moderating variables, and control variables, with detailed measurement methods and descriptions as follows:

Table 1
Variable definitions.

Variable Type	Name	Symbol	Definition	Description
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Dependent variables	Green Innovation Level (core indicator)	<i>Greenpatent</i>	$\ln(\text{Number of independently applied green invention patents in the current year} + \text{Number of independently applied green utility model patents in the current year} + 1)$	Measures the overall level of corporate green innovation activities. Following prior literature, this composite indicator captures both incremental and breakthrough innovations.
Independent variables	Policy Shock	<i>Treat</i> × <i>Post</i>	Multi-period DID interaction term, taking the value of 1 if the firm's registered place is in a green finance pilot zone and the year is in or after the establishment year of the pilot zone, otherwise 0	Green finance pilot zones were established in three batches (2017, 2019, 2022), capturing the net policy effect
Mediating variables	Financing Constraints	<i>SA index</i>	$SA\ index = -0.737 \times Size + 0.043 \times Size^2 - 0.040 \times Age$	The smaller the SA index, the stronger the financing constraints.
	Agency Costs	<i>Mfee</i>	Management Expense Ratio = Management expenses / Operating income	The higher the value, the higher the agency costs
	External Supervision	<i>Analyst attention</i>	Number of analysts following the firm	Reflecting the degree of analyst attention to enterprises
Moderating variables	State-owned Enterprise Dummy	<i>SOE</i>	Taking the value of 1 for state-owned enterprises and 0 for non-state-owned enterprises	Distinguishing the ownership attribute of enterprises
	Heavy Polluting Industries	<i>Pollute_1</i>	Taking the value of 1 for high-pollution industries and 0 for non-high-pollution industries (based on the "Catalogue for Classification Management of Environmental Protection Verification of Listed Companies")	Distinguishing the sensitivity of enterprises to environmental regulation
	Company Size	<i>Size</i>	$\ln(\text{Year-end total assets})$; when grouping, using the industry average as the boundary, taking the value of 1 for large enterprises (above the average) and 0 for small enterprises (below the average)	Reflecting the operating scale of enterprises
	High-tech Industry	<i>HighTech</i>	Taking the value of 1 for high-tech industries and 0 for non-high-tech industries (based on the CSRC industry	Distinguishing the technological intensity of enterprises

classification)				
Control variables	Financial Leverage	<i>Lev</i>	Total liabilities / Total assets	Reflecting the capital structure of enterprises
	Net Profit Margin On Total Assets	<i>ROA</i>	Net profit / Total assets	Reflecting the profitability of enterprises
	Largest Shareholder Ownership Ratio	<i>Top1</i>	Number of shares held by the largest shareholder / Total share capital	Reflecting the concentration of ownership
	Institutional Ownership Ratio	<i>Inst</i>	Number of shares held by institutional investors / Total share capital	Reflecting the participation degree of institutional investors
	Management Ownership Ratio	<i>Mshare</i>	Number of shares held by management / Total share capital	Reflecting the level of management incentives
	Proportion Of Independent Directors	<i>Indep</i>	Number of independent directors / Total number of board members	Reflecting the level of corporate governance
	Intensity Of Market Competition	<i>HHI</i>	Sum of squares of market shares of enterprises in the industry	The higher the HHI value, the weaker the industry competition.
	Ownership Balance Degree	<i>Balance2</i>	Sum of ownership ratios of the 2nd to 5th largest shareholders / Ownership ratio of the largest shareholder	The higher the value, the stronger the ownership balance capacity
	Firm Age	<i>Age</i>	$\ln(\text{Current year} - \text{Establishment year} + 1)$	Reflecting the operating years of enterprises

3.3. Sample and data sources

Due to data availability constraints for 2024 and to ensure consistency with the green finance policy context, we selected an initial sample of all A-share listed companies in China from 2000 to 2024. To construct a clean data set suitable for causal inference, we applied the following screening criteria: (1) excluding financial firms due to their unique accounting standards; (2) removing firms labeled ST, *ST, or PT during the sample period; (3) dropping observations with asset–liability

ratios >1 ; (4) setting the core sample period as 2012–2024, as the 2012 "Green Credit Guidelines" marked a key advancement in China's green finance policies; (5) excluding firms listed after 2012 to mitigate survivor bias and form a relatively balanced panel; (6) winsorizing all continuous variables at the 1% and 99% levels; and (7) applying listwise deletion for missing values. After screening, we obtained an unbalanced panel of 20,322 province–year observations, covering around 2,000 firms over 13 years, supporting multi-period DID estimation.

Green patent data are from the CNRDS platform. Following the IPC Green Inventory published by the World Intellectual Property Organization (WIPO), we matched the patent data to identify green patents. Firm-level financial and governance data are from CSMAR, and city-level control variables from the China City Statistical Yearbook. Indicators such as financing constraints and agency costs are derived from CSMAR, supplemented by the Wind database. Green finance pilot zone information is manually compiled from official policy documents issued by multiple national ministries.

All continuous variables are winsorized at the 1% level. A multi-period DID approach with firm and year fixed effects is employed to address endogeneity.

4. Empirical results

4.1. Descriptive statistics

Table 2 presents summary statistics for the main variables. The mean value of the dependent variable, *Greenpatent*, is 0.402, with a standard deviation of 0.846, indicating substantial variation in green innovation across sampled firms. The relatively low mean suggests that green innovation activities remain limited for most enterprises. The policy interaction term (*Treat*×*Post*) has a mean of 0.123, implying that approximately 12.3% of observations are treated by the GFPZ policy.

Regarding the mediating variables, the average *SA index* is -3.971, indicating that sampled firms face, on average, moderate financing constraints. The mean *Mfee* is 0.110, reflecting cross-firm heterogeneity in agency costs. Notably, *Analyst attention* exhibits considerable variation (SD=0.246), suggesting substantial differences in external scrutiny across firms.

Turning to the control variables, the means of *Lev* (0.413), *ROA* (0.039), and *Top1* (0.333) are consistent with the typical characteristics of A-share listed companies. The summary statistics for *Inst*, *Mshare*, *Indep*, *HHI*, and *Balance2* further capture heterogeneity in ownership structure, corporate governance, and industry competition.

Table 2
Descriptive statistics.

<i>Varname</i>	Obs	Mean	SD	Min	Median	Max
<i>Greenpatent</i>	20322	0.402	0.846	0.000	0.000	6.848
<i>Treat×Post</i>	20322	0.123	0.329	0.000	0.000	1.000
<i>SA index</i>	20322	-3.971	0.246	-4.892	-3.956	-3.215
<i>Mfee</i>	20322	0.110	0.085	0.012	0.089	0.892
<i>Analyst attention</i>	20322	0.417	0.246	0.000	0.432	0.937
<i>Lev</i>	20322	0.413	0.195	0.025	0.405	0.941
<i>ROA</i>	20322	0.039	0.061	-0.581	0.037	0.286
<i>Top1</i>	20322	0.333	0.147	0.044	0.311	0.815
<i>Inst</i>	20322	0.417	0.246	0.000	0.432	0.937
<i>Mshare</i>	20322	0.130	0.186	0.000	0.012	0.748
<i>Indep</i>	20322	0.377	0.056	0.250	0.364	0.625
<i>HHI</i>	20322	0.085	0.095	0.013	0.059	0.889
<i>Balance2</i>	20322	0.722	0.612	0.016	0.538	3.350

4.2. Baseline regression analysis

Table 3 reports the baseline estimates from Eq. (1), examining the impact of GFPZ

policy on corporate green innovation. Columns (1)–(5) progressively introduce control variables and fixed effects to assess the robustness of the policy effect. Column (1) includes only the policy interaction term ($Treat \times Post$) without controls; Columns (2)–(5) sequentially add firm-level controls, province fixed effects, and year fixed effects. All specifications control for province and year fixed effects to absorb time-invariant regional heterogeneity and common macroeconomic shocks. Standard errors are clustered at the industry level to account for within-industry correlation.

The results in [Table 3](#) show that the coefficient of $Treat \times Post$ is positive and statistically significant across all specifications. In Column (1), the coefficient is 0.098 ($p < 0.01$), indicating that the GFPZ policy promotes corporate green innovation without controls. As controls and fixed effects are progressively added, the coefficient remains positive and significant. In the fully saturated specification with province and year fixed effects (Column 5), the coefficient is 0.051 ($t=3.38$, $p < 0.01$). This finding suggests that, even after accounting for time-invariant regional heterogeneity and common annual shocks, the GFPZ policy exerts a significant positive effect on corporate green innovation, supporting [Hypothesis H1](#).

Turning to the control variables, Lev enters with a positive and significant coefficient, suggesting that moderate financial leverage may provide necessary funding for green innovation. ROA is positive but insignificant, possibly because highly profitable firms prioritize traditional profit-generating projects over green innovation investments. $Top1$ is negative and significant, indicating that excessive ownership concentration may impede green innovation. In contrast, $Inst$ is positive and significant, suggesting that institutional investors facilitate corporate green innovation. HHI is negative and significant, implying that stronger industry competition (i.e., lower HHI) encourages firms to engage more actively in green

innovation.

While the baseline results in Table 3 show that the GFPZ policy significantly promotes the overall level of corporate green innovation (*Greenpatent*), this composite measure may mask heterogeneous effects on different types of green innovation. Specifically, green invention patents represent high-quality, breakthrough innovations, whereas green utility model patents represent incremental, low-quality innovations. To uncover the differential impacts on innovation quality versus quantity, we conduct further analysis in Section 4.6 by disaggregating the composite measure.

Table 3
Baseline regression.

VARIABLES	(1)	(2)	(3)	(4)	(6)
	<i>Greenpatent</i>				
<i>Treat</i> × <i>Post</i>	0.098***(0.38)	0.083***(5.89)	0.055***(3.61)	0.125***(9.53)	0.051***(3.38)
<i>Lev</i>		0.662***(19.46)	0.652***(19.17)	0.144***(3.68)	0.102***(2.60)
<i>ROA</i>		0.769***(7.18)	0.762***(7.09)	0.125(1.63)	0.106(1.39)
<i>Top1</i>		-0.475***(-6.05)	-0.465***(-5.93)	-0.204**(-2.20)	0.032(0.34)
<i>Inst</i>		0.426***(9.08)	0.441***(9.39)	0.101**(1.99)	0.083*(1.65)
<i>Mshare</i>		0.374***(7.03)	0.406***(7.56)	-0.044(-0.84)	0.059(1.11)
<i>Indep</i>		0.435***(4.16)	0.398***(3.81)	0.163(1.60)	0.080(0.79)
<i>HHI</i>		-0.617***(-9.90)	-0.584***(-9.35)	-0.301***(-3.81)	-0.168**(-2.12)
<i>Balance2</i>		-0.017(-1.14)	-0.020(-1.29)	-0.013(-0.77)	-0.002(-0.09)
<i>Constant</i>		-0.055(-1.12)	-0.048(-0.96)	0.318*** (5.68)	0.278*** (4.97)
<i>Province Fixed Effects</i>	NO	NO	NO	YES	YES
<i>Year Fixed Effects</i>	NO	YES	YES	NO	YES
<i>Observations</i>	20322	20322	20322	20322	20322
<i>R</i> ²	0.033	0.038	0.037	0.701	0.707
<i>Adjusted R</i> ²	0.033	0.037	0.036	0.674	0.680

Notes: T-statistics are reported in parentheses; *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the industry level. Columns (1)–(5) sequentially add control variables and fixed effects as indicated.*

4.3. Robustness tests

To ensure the reliability of the main findings in the study, we conducted a series of robustness checks as follows:

4.3.1. Placebo tests

A key identifying assumption of the DID approach is that the establishment of GFPZs is the only factor driving the observed change in corporate green innovation around the policy implementation years. However, unobservable or omitted variables—such as concurrent environmental regulations or macroeconomic shocks—could confound the baseline estimates. To rule out this possibility, we conduct placebo tests, a widely used approach to detect potential biases from unobserved factors.

We design two distinct placebo tests. In each test, we artificially construct a “counterfeit” policy variable and re-estimate the baseline model. If the estimated coefficient of the counterfeit variable is statistically indistinguishable from zero, it suggests that the genuine policy effect is not driven by spurious factors.

(1) *Time-based placebo test.* To assess whether the estimated policy effect is driven by random timing rather than actual policy implementation, we conduct a time-based placebo test. Keeping the treatment group composition unchanged, we randomly reassign policy implementation years to treated firms, generating a placebo time indicator $Post_{i,t}^{fake}$. This placebo indicator is then interacted with the actual treatment dummy $Treat_i$ to construct a counterfeit policy term $Treat_i \times Post_{i,t}^{fake}$, and we re-estimate the baseline model:

$$Greenpatent_{i,t} = \alpha_0 + \alpha_1 Treat_i \times Post_{i,t}^{fake} + \sum_k \beta_k Controls_{k,i,t} + \gamma_i + \mu_t + \varepsilon_{i,t} \quad (2)$$

We repeat this procedure 500 times, each with a different random assignment of policy years. [Fig.1\(a\)](#) plots the kernel density of the 500 estimated α_1 coefficients; the vertical dashed line indicates the true baseline coefficient (0.051). The placebo coefficients are centered around zero, and the true estimate lies far outside the main mass of the distribution, suggesting that the baseline result is unlikely to be driven by random time-specific factors.

(2) *Region-based placebo test.* To further rule out the possibility that unobserved regional characteristics are driving the results, we perform a region-based placebo test. We randomly select a set of firms to serve as a “fake treatment group” by randomly assigning a placebo treatment status $Treat_i^{fake}$. This fake treatment dummy is then interacted with the actual policy timing $Post_{i,t}$ to create a counterfeit policy term $Treat_i^{fake} \times Post_{i,t}$. The regression model is specified as:

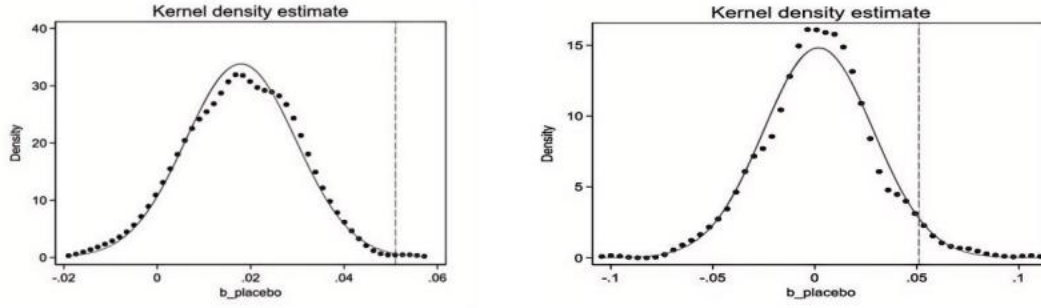
$$Greenpatent_{i,t} = \alpha_0 + \alpha_1 Treat_i^{fake} \times Post_{i,t} + \sum_k \beta_k Controls_{k,i,t} + \gamma_i + \mu_t + \varepsilon_{i,t} \quad (3)$$

We repeat this random assignment and estimation 500 times. [Fig.1\(b\)](#) displays the kernel density of the 500 estimated coefficients. Again, the distribution of placebo coefficients is tightly centered around zero, while the true baseline coefficient (0.051, indicated by the dashed line) falls well outside this range. This confirms that the observed policy effect is not attributable to accidental regional factors.

Together, the results of both placebo tests provide strong evidence that the positive effect of green finance pilot zones on corporate green innovation identified in the baseline regression is robust and not driven by unobservable or random factors.

(a) :Randomly Permuted Policy Years

(b):Randomly Permuted Treatment Group Provinces

**Fig. 1.** Kernel Density of Placebo Coefficients (500 Times)

4.3.2. PSM-DID analysis

To further verify the robustness of the baseline results, this section employs the Propensity Score Matching Difference-in-Differences (PSM-DID) method to alleviate potential selection bias and endogeneity concerns. First, a logit model is used to estimate the propensity score for each enterprise, which represents the probability of being in the treatment group:

$$P(Treat_i=1|X_i)=\text{Logit}\left(\gamma_0+\sum_k \gamma_k Controls_{k,i}\right) \quad (4)$$

where X_i denotes a vector of matching covariates, including all control variables from the baseline regression, and $Treat_i$ is the group identifier, defined consistently with Eq. (1). Subsequently, we perform 1:1 nearest neighbor matching without replacement to construct a matched sample. After ensuring that the parallel trend assumption and covariate balance are satisfied, the standard DID model is re-estimated on this matched sample. The estimation results are reported in Table 4. The coefficient of the core explanatory variable $Treat \times Post$ is 0.087, with a t-statistic of 4.97, which is significant at the 1% level. This positive and statistically significant coefficient is consistent with the baseline findings, confirming that the policy effect remains robust after accounting for observable differences between the treatment and

control groups.

Table 4
Robustness Tests: PSM-DID Estimation.

<i>VARIABLES</i>	PSM-DID
<i>Treat×Post</i>	0.087***(4.97)
Control variables	YES
Province fixed effects	YES
Year fixed effects	YES
Observations	17858
R ²	0.707
Adjusted R ²	0.680

Note: T-statistics are reported in parentheses; *** p<0.01, ** p<0.05, * p<0.1.

Taken together, the placebo tests and the PSM-DID estimation provide strong evidence that the baseline findings are not driven by unobservable con-founders or selection bias. The policy effect of green finance pilot zones on corporate green innovation is therefore robust and credible.

4.4. Mechanism analysis

To uncover the underlying channels through which the GFPZ policy affects corporate green innovation, we conduct a mediation analysis following the stepwise regression approach. Three potential mediators are examined: financing constraints, agency costs, and external supervision. The testing procedure involves two steps: first, we estimate the effect of the policy on each mediator; second, we include both the policy term and the mediator in the baseline model to observe changes in the policy coefficient. The corresponding models are specified as follows:

$$M_{i,t} = \beta_0 + \beta_1 Treat_i \times Post_{i,t} + \sum_k \beta_k Controls_{k,i,t} + \gamma_i + \mu_t + \varepsilon_{i,t} \quad (5)$$

$$Greenpatent_{i,t} = \gamma_0 + \gamma_1 Treat_i \times Post_{i,t} + \gamma_2 M_{i,t} + \sum_k \gamma_k Controls_{k,i,t} + \gamma_i + \mu_t + \varepsilon_{i,t} \quad (6)$$

where $M_{i,t}$ represents the mediating variable (i.e., financing constraints, agency costs, or external supervision), and all other variables are defined consistently with Eq. (1). A mediating effect is established if β_1 in Eq. (5) is significant and γ_2 in Eq. (6) is significant, while the coefficient γ_1 becomes smaller in magnitude or less significant compared to the baseline estimate. $Treat_i \times Post_{i,t}$, $Controls_{k,i,t}$, γ_i , μ_t , and $\varepsilon_{i,t}$ have the same definitions as in Eq. (1).

4.4.1. Financing constraint channel

We first employ the *SA index* to measure firms' financing constraints. A higher SA index indicates looser financing constraints. Column (1) of Table 5 reports the estimates of Eq. (5) with the *SA index* as the dependent variable. The coefficient of $Treat \times Post$ is 0.009 ($p < 0.01$), indicating that the GFPZ policy significantly alleviates firms' financing constraints. Column (2) presents the results of Eq. (6), including both the policy term and the *SA index*. The coefficient of $Treat \times Post$ drops from the baseline 0.051 to 0.046 ($t = 3.04$), while the *SA index* itself is positively associated with green innovation (coefficient=0.566, $t=8.27$). These findings suggest that financing constraints serve as a partial mediator, supporting Hypothesis 1.

4.4.2. Agency cost channel

We use the management expense ratio (*Mfee*) as a proxy for agency costs, where a higher value reflects greater agency problems. Column (3) of Table 5 displays the results of Eq.(5) with *Mfee* as the dependent variable. The coefficient of $Treat \times Post$ is

-0.004, significant at the 5% level, implying that the GFPZ policy reduces agency costs. In column (4), when *Mfee* is added to the green innovation regression, the coefficient of *Treat*×*Post* declines to 0.050 (*t*=3.32) and *Mfee* is negatively significant (coefficient = -0.261, *t*=-3.44). This pattern supports the mediating role of agency costs, in line with [Hypothesis 1](#).

4.4.3. External supervision channel

We measure external supervision by *Analyst attention*, standardized by industry. Column (5) of [Table 5](#) reports the estimation of [Eq.\(5\)](#) with *Analyst* as the dependent variable. The coefficient of *Treat*×*Post* is 0.048, significant at the 5% level, indicating that the GFPZ policy attracts more analyst coverage. Column (6) incorporates *Analyst* into the green innovation model. The coefficient of *Treat*×*Post* decreases to 0.049 (*t*=3.25) and *Analyst* exerts a positive and significant effect (coefficient = 0.040, *t*=6.78). Thus, external supervision partially mediates the policy effect, further corroborating [Hypothesis 1](#).

Table 5
Mechanism Test: The Mediating Effects of Financing Constraints, Agency Costs, and External Supervision.

	(1)	(2)	(3)	(4)	(5)	(6)
VARIABLES	<i>SA index</i>	<i>Greenpatent</i>	<i>Mfee</i>	<i>Greenpatent</i>	<i>Analyst attention</i>	<i>Greenpatent</i>
<i>Treat</i> × <i>Post</i>	0.009*** (5.58)	0.046*** (3.04)	-0.004** (-2.40)	0.050*** (3.32)	0.048** (2.53)	0.049*** (3.25)
<i>SA index</i>		0.566*** (8.27)				
<i>Mfee</i>				-0.261*** (-3.44)		
<i>Analyst attention</i>						0.040***

						(6.78)
<i>Lev</i>	-0.041*** (-9.90)	0.125*** (3.19)	-0.065*** (-17.28)	0.085** (2.14)	0.454*** (9.26)	0.084** (2.13)
<i>ROA</i>	-0.005 (-0.61)	0.109 (1.43)	-0.229*** (-30.96)	0.046 (0.59)	3.027*** (31.55)	-0.014 (-0.17)
<i>Top1</i>	0.123*** (12.07)	-0.037 (-0.39)	0.036*** (3.95)	0.042 (0.44)	-2.381*** (-19.94)	0.126 (1.32)
<i>Inst</i>	0.054*** (10.11)	0.052 (1.03)	-0.017*** (-3.55)	0.078 (1.56)	2.130*** (33.79)	-0.001 (-0.03)
<i>Mshare</i>	0.137*** (24.20)	-0.019 (-0.35)	0.013** (2.57)	0.062 (1.18)	1.154*** (17.36)	0.013 (0.25)
<i>Indep</i>	0.004 (0.38)	0.077 (0.77)	0.012 (1.21)	0.083 (0.82)	0.287** (2.27)	0.068 (0.68)
<i>HHI</i>	0.000 (0.04)	-0.168** (-2.13)	-0.027*** (-3.50)	-0.175** (-2.21)	0.188* (1.89)	-0.176** (-2.22)
<i>Balance2</i>	0.020*** (11.29)	-0.013 (-0.77)	0.005*** (3.19)	-0.000 (-0.01)	-0.276*** (-12.97)	0.009 (0.55)
<i>Constant</i>	-3.971*** (-662.71)	2.525*** (9.11)	0.110*** (20.36)	0.307*** (5.42)	-0.456*** (-6.49)	0.296*** (5.29)
Province fixed effects	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES
Observations	20322	20322	20322	20322	20322	20322
R ²	0.971	0.708	0.670	0.707	0.668	0.707
Adjusted R ²	0.968	0.681	0.641	0.680	0.638	0.681

Note: T-statistics are reported in parentheses; *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the industry level. Columns (1), (3), and (5) report the effect of the policy on each mediator. Columns (2), (4), and (6) include both the policy term and the mediator in the green innovation regression.

4.5. Heterogeneity analysis

To further investigate the differential impacts of the green finance pilot zone policy on the green innovation behaviors of enterprises with distinct characteristics, this study introduces interaction terms between the policy dummy variable and a series of firm-level moderating variables into the baseline model. The moderating variables include ownership nature (*SOE*), pollution attribute (*Pollute_1*), firm size (*Size*), and technological attribute (*HighTech*). The specific model is specified as follows:

$$Greenpatent_{i,t} = \delta_0 + \delta_1 Treat_i \times Post_{i,t} + \delta_2 Treat_i \times Post_{i,t} \times Moderator_{i,t} + \delta_3 Moderator_{i,t} + \sum_k \delta_k Controls_{k,i,t} + \gamma_i + \mu_t + \varepsilon_{i,t} \quad (7)$$

Where $Moderator_{i,t}$ is the moderating variable, including ownership nature (*SOE*), pollution attribute (*Pollute_1*), province size (*Size*), and technological attribute (*HighTech*); $Treat_i \times Post_{i,t}$, $Controls_{k,i,t}$, γ_i , μ_t , and $\varepsilon_{i,t}$ have the same definitions as in Eq. (1); the core focus is on the coefficient δ_2 of the interaction term, which reflects the moderating effect.

Here, $Moderator_{i,t}$ sequentially represents the four moderating variables mentioned above. $Treat_i \times Post_{i,t}$, $Controls_{k,i,t}$, γ_i , μ_t , and $\varepsilon_{i,t}$ are defined identically to those in Eq.(1). Our primary focus is on the coefficient δ_2 of the triple interaction term, which captures the heterogeneity in the policy effect across different firm types.

Table 6 presents the regression results of the heterogeneity analysis. Regarding the moderating effect of ownership nature, the coefficient of the interaction term $Treat \times Post \times SOE$ in column (2) is $-0.063(t=-2.15)$, significantly negative at the 5% level. This indicates that state ownership significantly weakens the promotional effect of the policy on green innovation, implying that the policy effect is more pronounced in non-state-owned enterprises (*Non-SOEs*). A plausible explanation is

that SOEs typically benefit from substantial government subsidies and preferential credit access. The resultant soft budget constraints may render them less responsive to market-oriented incentives like green finance policies. Conversely, Non-SOEs, facing fiercer market competition and tighter financing constraints, are more motivated to leverage policy dividends for green transformation, thereby exhibiting a more significant innovation boost.

Concerning the moderating effect of pollution attributes, the coefficient of the interaction term $Treat \times Post \times Pollute_1$ in column (4) is -0.229 ($t=-4.32$), significant at the 1% level. This suggests that being a high-pollution firm dampens the policy effect, meaning the policy's promotional impact on green innovation is stronger for non-high-pollution enterprises. This result may be linked to the technology lock-in effect prevalent in high-pollution industries. Long-term reliance on energy-intensive and high-emission production models makes their green technology transition costly and time-consuming, hindering short-term responses to policy incentives. Furthermore, green finance policies often impose stricter credit constraints on high-polluting sectors, further limiting their access to innovation resources.

In terms of firm size, the coefficient of the interaction term $Treat \times Post \times Size$ in column (6) is 0.052 ($t=2.31$), significantly positive at the 5% level. This demonstrates that larger enterprises amplify the policy's positive impact on green innovation. This is likely because large firms possess more abundant internal funds, more established R&D systems, and greater risk-bearing capacity, enabling them to more effectively translate policy preferences into tangible green R&D investments. While small enterprises have urgent financing needs, their limited collateral and lower information transparency constrain their ability to benefit from green finance schemes.

Finally, regarding technological attributes, the coefficient of the interaction

term $Treat \times Post \times HighTech$ in column (8) is 0.075 ($t=2.89$), significant at the 1% level. This indicates that high-tech enterprises strengthen the policy's promotional effect. High-tech firms typically have higher R&D intensity and better technological absorption capabilities, making them more adept at assimilating and generating green technologies. Consequently, they respond more sensitively to green finance policies and are more likely to achieve high-quality green innovation outputs.

In summary, the heterogeneity analysis reveals that the innovation-promoting effect of the green finance pilot zone policy varies significantly based on firm ownership, pollution intensity, size, and technological characteristics. The positive impact is more pronounced for non-SOEs, non-heavy polluters, large enterprises, and high-tech firms. These findings not only further validate research [Hypothesis 1](#) but also provide empirical evidence supporting the implementation of differentiated and targeted policy interventions.

Table 6
Heterogeneity analysis results.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>VARIABLES</i>	SOEs	Non-SOEs	High-pollution	Non-high-pollution	Large enterprises	Small enterprises	High-tech	Non-high-tech
$Treat \times Post$	0.019 (0.68)	0.082*** (4.34)	-0.131*** (-3.88)	0.098*** (5.79)	0.070*** (3.11)	0.018 (0.90)	0.078** * (3.64)	0.003 (0.14)
$Treat \times Post \times SOE$		-0.063** (-2.15)						
$Treat \times Post \times Pollute_1$				-0.229*** (-4.32)				
$Treat \times Post \times Size$						0.052** (2.31)		
$Treat \times Post \times HighTech$								0.075*** (2.89)
<i>Lev</i>	0.187*** (2.65)	0.104** (2.10)	0.010 (0.12)	0.142*** (3.18)	0.212*** (3.29)	0.053 (1.11)	0.111* (1.94)	0.065 (1.28)
<i>ROA</i>	0.150 (0.91)	0.069 (0.78)	0.461** (2.50)	0.050 (0.59)	0.153 (1.01)	0.066 (0.79)	0.080 (0.78)	0.114 (1.04)
<i>Top1</i>	-0.310* (-1.71)	0.108 (0.44)	-0.292 (-1.18)	0.156 (0.63)	-0.089 (-0.37)	0.124 (0.52)	0.226 (0.94)	-0.104 (-0.42)

	(-1.74)	(0.87)	(-1.48)	(1.42)	(-0.65)	(0.93)	(1.60)	(-0.84)
<i>Inst</i>	-0.051 (-0.45)	0.101* (1.70)	0.120 (1.09)	0.075 (1.32)	0.098 (1.20)	0.006 (0.10)	0.146** (2.10)	0.045 (0.65)
<i>Mshare</i>	-0.463 (-1.24)	-0.015 (-0.25)	0.302** (2.43)	0.023 (0.39)	0.077 (0.74)	-0.058 (-0.97)	0.117 (1.63)	0.016 (0.21)
<i>Indep</i>	0.148 (0.95)	0.007 (0.05)	-0.033 (-0.15)	0.067 (0.58)	-0.023 (-0.14)	0.161 (1.26)	-0.077 (-0.53)	0.298** (2.33)
<i>HHI</i>	-0.320*** (-2.79)	0.010 (0.08)	-0.139 (-0.46)	-0.134 (-1.59)	- 0.457*** (-3.63)	0.051 (0.52)	-0.326* (-1.75)	-0.144* (-1.81)
<i>Balance2</i>	-0.005 (-0.13)	-0.006 (-0.29)	-0.024 (-0.67)	0.006 (0.33)	-0.020 (-0.73)	0.000 (0.01)	0.022 (0.93)	-0.017 (-0.73)
<i>Constant</i>	0.462*** (4.85)	0.263*** (3.60)	0.403*** (3.38)	0.231*** (3.59)	0.424*** (4.85)	0.179** (2.47)	0.342** * (4.18)	0.121 (1.64)
Province fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
Year fixed effects	YES	YES	YES	YES	YES	YES	YES	YES
Observations	7091	13215	4455	15856	10154	10168	12365	7943
R ²	0.760	0.681	0.595	0.731	0.746	0.606	0.713	0.667
Adjusted R ²	0.742	0.663	0.571	0.714	0.728	0.587	0.695	0.648

Notes: T-statistics are reported in parentheses; *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the industry level. Each column reports regression results for a specific subsample as indicated in the column header.*

4.6. Differential Effects on Innovation Quality vs. Quantity

The baseline analysis in [Section 4.2](#) establishes that the GFPZ policy significantly promotes the overall level of corporate green innovation. However, a critical question remains: does this reflect genuine improvements in innovation quality, or merely an expansion in low-quality, incremental innovations? Given that green utility model patents are characterized by lower authorization thresholds and shorter commercialization cycles, they serve as intuitive indicators of innovation quantity. Conversely, green invention patents require higher inventiveness, involve greater R&D investment, and yield higher-value outcomes, thus constituting the core basis

for measuring innovation quality.

To address this question, we disaggregate the composite green innovation measure into its two components: green invention patents (representing high-quality innovation, denoted as *Green Innovation 3*) and green utility model patents (representing low-quality innovation, denoted as *Green Innovation 2*). We re-estimate the baseline model using these disaggregated measures as dependent variables:

$$Greenpatent_{i,t}^q = \alpha_0 + \alpha_1 Treat_i \times Post_{i,t} + \sum_k \beta_k Controls_{k,i,t} + \gamma_i + \mu_t + \varepsilon_{i,t} \quad (8)$$

where $Greenpatent_{i,t}^q$ represents either the quantitative dimension (*Green Innovation 2*) or the qualitative dimension (*Green Innovation 3*). All other variables are defined consistently with Eq. (1). Table 7 reports the estimation results.

Table 7

Joint Effects of Green Financial Policy on Green Innovation: Quantity vs. Quality Evidence.

VARIABLES	(1)	(2)
	<i>Green Innovation 2</i>	<i>Green Innovation 3</i>
<i>Treat</i> × <i>Post</i>	0.044***(3.45)	0.038***(3.21)
Control variables	YES	YES
Province fixed effects	YES	YES
Year fixed effects	YES	YES
Observations	20322	20322
R ²	0.695	0.636
Adjusted R ²	0.668	0.603

Notes: t-statistics are reported in parentheses; *** p<0.01, ** p<0.05, * p<0.1. Standard errors are clustered at the industry level. Column (1) uses green utility model patents as the dependent variable (quantity). Column (2) uses green invention

patents as the dependent variable (quality).

As shown in [Table 7](#), the coefficient of *Treat*×*Post* is positive and statistically significant at the 1% level in both specifications. Specifically, the policy shock yields coefficients of 0.044 for green utility model patents and 0.038 for green invention patents. The coefficient for utility models (quantity) is slightly larger than that for invention patents (quality), reflecting the shorter R&D cycles and lower technical thresholds associated with incremental innovations, which respond more quickly to policy incentives. Importantly, the significant positive effect on invention patents confirms that the GFPZ policy genuinely enhances green innovation quality, rather than merely inflating patent counts.

This pattern is consistent with the operational logic of the three mechanisms identified earlier. The alleviation of financing constraints provides direct financial support for enterprises to conduct low-threshold, rapid-implementation utility model innovations. The reduction in agency costs prevents the misappropriation of funds, ensuring stable investment in utility model innovation. The strengthening of external supervision curbs short-sighted behaviors that emphasize quantity over quality, compelling enterprises to simultaneously allocate resources to the R&D of high-value invention patents. The observed coefficient differences also reflect the inherent characteristics of innovation activities: invention patents require longer R&D cycles and entail higher technical risks, leading to a certain lag in the realization of policy effects, whereas utility model innovations respond more rapidly to policy incentives.

In summary, these findings provide an important supplement to [Hypothesis 1](#). The GFPZ policy does not merely encourage enterprises to engage in extensive innovation characterized by low quality and high quantity. Rather, through the three-dimensional synergistic mechanism of "financial support—internal governance—external

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